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Introduction

- **Project / Innovation Name:** ___Infinite EV Engine___
- **Team Name:** ___Innovate X___
- **Team Members and Classes:** ___Gurkamalveer Singh (8th) __and _Arshdeep Singh (8th)
- **School Name:** __ Satya Bharti Adarsh Sen. Sec. School Rauni __
- **Briefly introduce your idea in 1–2 sentence:-** Our project introduces an automated dual-battery power management and kinetic energy recovery system that extends electric vehicle operational range by alternating battery packs and harvesting rotational wheel energy. It uses an Arduino Uno to continuously monitor voltages and execute relay switching, ensuring uninterrupted vehicle mobility and improved energy efficiency.

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Problem Statement

- **What problem have you identified?**
- We identified EV range anxiety, long stationary charging times, and rapid battery degradation caused by continuous single-pack deep discharge cycles. This prevents light electric vehicles from achieving higher operational efficiency.
- **Who faces this problem?**
- EV drivers, electric rickshaw and commercial fleet operators, as well as light electric vehicle manufacturers, face frequent downtime and high battery replacement costs.
- **Why is it important to solve this problem?**
- Solving it extends vehicle uptime, reduces reliance on stationary grid charging, and mitigates cell degradation to make electric transport more sustainable and economical.
- **Real-life example:**
- For instance, an e-rickshaw driver operating for commercial transport often experiences sudden battery drain mid-trip, forcing hours of stationary charging that halts daily earnings.

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Proposed Solution

- **What is your proposed solution?**
- Our solution is an automated dual-battery power management and kinetic energy recovery system (KERS) that uses two switchable battery packs, coupled dynamos, and an Arduino Uno to optimize vehicle power.
- **How will it solve the problem?**
- It alternates power delivery between Battery A and Battery B while harvesting kinetic energy from wheel rotation via dynamos to trickle-charge the idle battery pack, ensuring uninterrupted operation.
- **What makes your idea useful or different?**
- Unlike single-pack systems, it actively recovers lost rotational energy, reduces continuous discharge thermal stress on batteries, and provides a low-cost, automated retrofit solution for light electric vehicles.

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Working and Description of the Idea

- **Working Mechanism (Step-by-Step):** Battery A powers the motor while dual dynamos harvest rotational energy to charge Battery B. An Arduino Uno continuously monitors their voltages via sensors. When Battery A drops, it triggers a relay to automatically switch loads for continuous operation.

- **Materials & Technology:** We use an Arduino Uno microcontroller, dual 12V batteries, DC dynamos, a 5V relay module, voltage sensors, and a 16x2 LCD display. The system is programmed in Embedded C++ using Arduino frameworks.

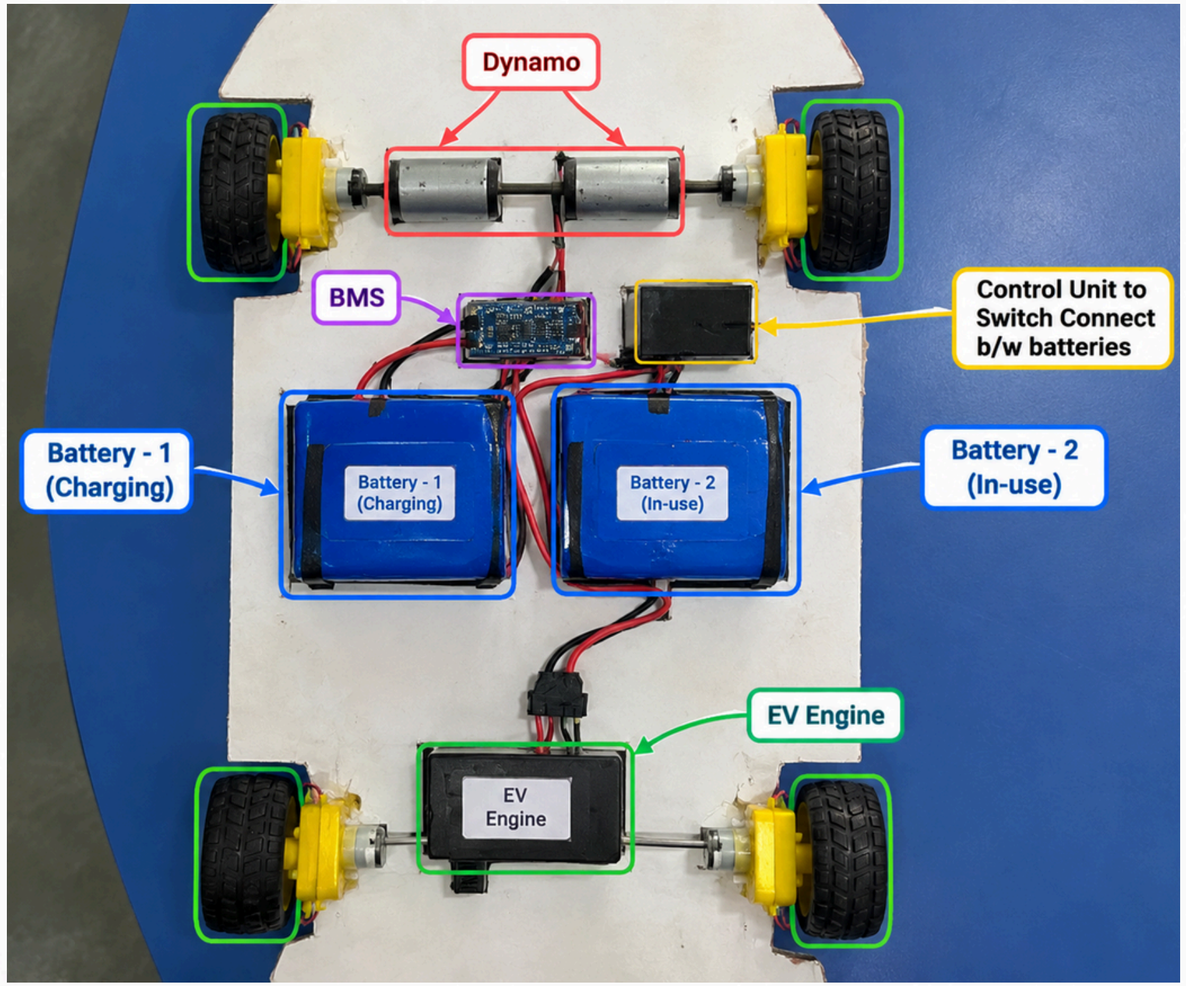
- **System Flowchart:**

Wheel Motion -----> Dynamos -------> Charging Battery B

Arduino Uno Monitors Voltages ------> Relay Module Switches Loads

LCD Screen Displays Live Telemetry

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Conclusion

- **What benefits will your innovation provide and who will benefit?**
- Our innovation provides active range extension, reduced battery cell thermal stress, and lower grid-charging frequency. The direct beneficiaries are EV drivers, commercial e-rickshaw fleet operators, and light electric vehicle manufacturers.

- **What impact could it have in the future?**
- In the future, it can significantly scale up operational uptime for commercial light electric transport. Future iterations can integrate advanced Maximum Power Point Tracking (MPPT) charge controllers and IoT cloud analytics for predictive battery health management.

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Thank you